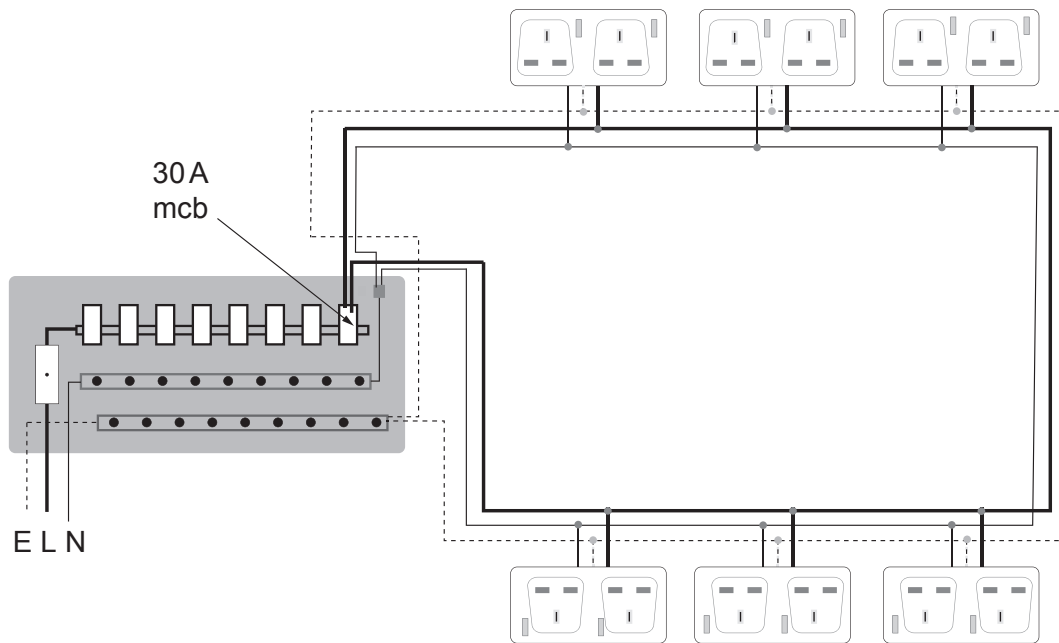


2. The diagram shows a common domestic ring main circuit.



(a) Give **two** advantages of having a ring main circuit in a house.

[2]

.....

.....

.....

.....



- (b) (i) State the function of the live wire. [1]

.....

.....

- (ii) Describe the function of the earth wire. [2]

.....

.....

.....

- (c) The following information is printed on the bottom of a domestic kettle:

TYPE: KBZ3001.BQ
230 V ~ a.c.
2760 W
CAPACITY 1.5L
SERIAL NUMBER 53023

- (i) State the differences between alternating current (a.c.) and direct current (d.c.). [2]

.....

.....

.....

- (ii) The voltage of the mains electricity supply in the U.K. is 230V. Use equations from page 2 and information on the kettle to calculate the resistance of the kettle heater. [4]

Resistance = Ω

